Efficient Memory Management for Large Language Model Serving with PagedAttention

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Abstract

This paper investigates architectures with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Efficient Memory Management for Large Language Model Serving with PagedAttention.

2. Technical Methodology & Empirical Findings

Section topic: Architectures
Question context: How does KV caching reduce computational redundancy during autoregressive token generation?
Key Finding: KV caching retains keys and values from preceding time steps in GPU memory, avoiding redundant re-computation of historical states at each sequential decoding step [PagedAttention, p. 3].
Evidence passage: Stores key and value vectors of previously generated tokens in memory Avoids recomputing attention states for historical prompt and generation tokens

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Efficient Memory Management for Large Language Model Serving with PagedAttention. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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